Highest-weight norm extension criterion for general G-zips (Theorem 4 / Proposition 3.9)

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Editorial scope. One entire norm-extension iff criterion and complete current proof; no mathematical repair or formal admission.

Allowed standard prerequisites (editorial).

- Zip orbit parametrization and finite stabilizer. [PWZ11] Theorems 7.5, 11.2, 11.3, 11.5; [KW18] Lemma 3.2.1; [IK21] Corollary 3.2.3.. Original Section 2 explicitly states orbit and invariant descriptions; these are earlier external results.
- Root-curve boundary test and Frobenius factorization. [IK21] Automorphic vector bundles on the stack of G-zips, Forum Math. Sigma 9 (2021), e37, DOI 10.1017/fms.2021.32, Section 3.1 and Proposition 3.1.4; [Kos19] Section 3.2.. Earlier geometric extension test and root-curve factorization are explicitly attributed at source lines 1043, 1059, 1070 and 1082. The current general-parabolic valuation computation is fully retained, not replaced by this citation.
- Earlier finite-field-parabolic argument and Bruhat counting. [Kos19] Automorphic Forms on the Stack of G-Zips, Results Math. 74 (2019), 91, DOI 10.1007/s00025-019-1021-z, Theorem 2, Theorem 3.5.3, Lemmas 3.4.4–3.4.5 and Proposition 3.5.2.. Author explicitly distinguishes the earlier finite-field-parabolic scope. Double-coset valuation invariance and current Frobenius orbit sum are reproduced through line 1125; finite Bruhat cardinalities remain cited tools.
- Reductive-group realizations and induced representations. [Spr98] Sections 8.1.4 and proof of Lemma 9.2.2; [Jan03] Section 5.8; [IK24] Proposition 3.2.1.. Standard root SL2 maps, highest-weight induced modules and flag pushforward; definitions and the required identifications are retained in Section 2.

Source reference. Original proposition 3.8: persistence-modules/EPIGA15861/sources/author-v5.tex, source lines 1012–1024. Forward-looking contextual reference in Section 2, not an input to the norm criterion; preserved as exact original-source pointer.

Authors and source. Naoki Imai and Jean-Stefan Koskivirta, *Weights of mod p automorphic forms and partial Hasse invariants*, exact arXiv:2211.16207v5 (12 July 2026), DOI 10.46298/epiga.2026.15861. Wushi Goldring authored the separate appendix, not the selected norm criterion.
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Counted scope. One result: entire Theorem 4 / Proposition 3.9, the highest-weight norm extension criterion for general cocharacter data, including parabolics not defined over the finite field. Introductory statement, complete Section 2, norm definition and all boundary/root-curve, Frobenius, valuation and Bruhat arguments through the end of the proof are retained. Supporting statements are not additional counted results.
Changes and dependencies. Journal furniture is replaced by a standard wrapper; original section, theorem and displayed-equation numbering is retained. The earlier results of Pink–Wedhorn–Ziegler, Imai–Koskivirta and Koskivirta are attributed external inputs, not represented as reproduced proofs. The current argument after the author's "similar" remark is reproduced in full. No mathematical repair or new proof is supplied.

Preparation only. No formal problem number or mathematical certification. Saved in-flight identities end on 14 October 2026 at 17:36:45 UTC; later overlap is unconfirmed. AI assistance concerns selection, metadata and packaging; no assertion about author AI use is inferred.

Our final result is the construction of natural mod p automorphic forms attached to the highest-weight vectors of the representations $V_I(\lambda)$. Let λ be an L-dominant character, and let $f_\lambda \in V_I(\lambda)$ denote the highest-weight vector of $V_I(\lambda)$. There is a natural way of defining the norm $\mathbf{f}_\lambda := \mathrm{Norm}_{L_\varphi}(f_\lambda)$ of f_λ. Here L_φ is a certain finite (generally nonsmooth) subgroup of L containing $L_0(\mathbb{F}_q)$. There is an integer $m \geq 0$, determined by L_φ, such that the norm $\mathrm{Norm}_{L_\varphi}(f_\lambda)$ is a section of $\mathcal{V}_I(d\lambda)$ (where $d = q^m|L_0(\mathbb{F}_q)|$) over the μ-ordinary locus $\mathcal{U}_\mu$ of G-$\mathtt{Zip}^\mu$ (see Section 3.5 for details). For $\alpha \in \Delta$, let r_α be the smallest integer $r \geq 1$ such that $\sigma^r(\alpha) = \alpha$.

Theorem 4 (Proposition 3.9)**.** *The section $\mathbf{f}_\lambda$ extends to G-$\mathtt{Zip}^\mu$ if and only if for all $\alpha \in \Delta \setminus \Delta_L$ one has*

$$\sum_{w \in W_{L_0}(\mathbb{F}_q)} \sum_{i=0}^{r_\alpha - 1} q^{i+\ell(w)} \left\langle w\lambda, \sigma^i(\alpha^\vee) \right\rangle \leq 0.$$

2 Preliminaries and background on the stack of G-zips

2.1 Notation

Throughout the paper, p is a prime number, q is a power of p, and $\mathbb{F}_q$ is a finite field with q elements. We write $k = \overline{\mathbb{F}}_q$ for an algebraic closure of $\mathbb{F}_q$. The notation G will always denote a connected reductive group over $\mathbb{F}_q$. For a k-scheme X, we denote by $X^{(q)}$ its qth power Frobenius twist and by $\varphi\colon X \to X^{(q)}$ its relative Frobenius morphism. Write $\sigma \in \mathrm{Gal}(k/\mathbb{F}_q)$ for the q-power Frobenius. We will always write (B, T) for a Borel pair of G; *i.e.*, $T \subset B \subset G$ are a maximal torus and a Borel subgroup in G. We do not assume that T is split over $\mathbb{F}_q$. Let B^+ be the Borel subgroup of G opposite to B with respect to T (*i.e.*, the unique Borel subgroup B^+ of G such that $B^+ \cap B = T$). We will use the following notation:

- As usual, $X^*(T)$ (resp. $X_*(T)$) denotes the group of characters (resp. cocharacters) of T. The group $\mathrm{Gal}(k/\mathbb{F}_q)$ acts naturally on these groups. Let $W = W(G_k, T)$ be the Weyl group of G_k. Similarly, $\mathrm{Gal}(k/\mathbb{F}_q)$ acts on W. Furthermore, the actions of $\mathrm{Gal}(k/\mathbb{F}_q)$ and W on $X^*(T)$ and $X_*(T)$ are compatible in a natural sense. We write $W(\mathbb{F}_q)$ for the $\mathrm{Gal}(k/\mathbb{F}_q)$-fixed subgroup of W.
- We write $\Phi \subset X^*(T)$ for the set of T-roots of G.
- We write $\Phi^+ \subset \Phi$ for the system of positive roots with respect to B^+ (*i.e.*, $\alpha \in \Phi^+$ when the α-root group U_α is contained in B^+). This convention may differ from that of other authors. We use it to match the conventions of previous publications [GK19a, Kos19].
- We write $\Delta \subset \Phi^+$ for the set of simple roots.
- For $\alpha \in \Phi$, let $s_\alpha \in W$ be the corresponding reflection. The system $(W, \{s_\alpha \mid \alpha \in \Delta\})$ is a Coxeter system. We write $\ell\colon W \to \mathbb{N}$ for the length function. Hence $\ell(s_\alpha) = 1$ for all $\alpha \in \Delta$. Let w_0 denote the longest element of W.
- For a subset $K \subset \Delta$, let W_K denote the subgroup of W generated by $\{s_\alpha \mid \alpha \in K\}$. Write $w_{0,K}$ for the longest element in W_K.
- Let ${}^K W$ (resp. W^K) denote the subset of elements $w \in W$ that have minimal length in the coset $W_K w$ (resp. wW_K). Then ${}^K W$ (resp. W^K) is a set of representatives of $W_K \backslash W$ (resp. W/W_K). The map $g \mapsto g^{-1}$ induces a bijection ${}^K W \to W^K$. The longest element in the set ${}^K W$ is $w_{0,K} w_0$.
- Let $X^*_+(T)$ denote the set of dominant characters, *i.e.*, characters $\lambda \in X^*(T)$ such that $\langle \lambda, \alpha^\vee \rangle \geq 0$ for all $\alpha \in \Delta$.
- For a subset $I \subset \Delta$, let $X^*_{+,I}(T)$ denote the set of characters $\lambda \in X^*(T)$ such that $\langle \lambda, \alpha^\vee \rangle \geq 0$ for all $\alpha \in I$. We call them I-dominant characters.
- Let $P \subset G_k$ be a parabolic subgroup containing B, and let $L \subset P$ be the unique Levi subgroup of P containing T. Then we define a subset $I_P \subset \Delta$ as the unique subset such that $W(L, T) = W_{I_P}$. For an arbitrary parabolic subgroup $P \subset G_k$ containing T, we define $I_P \subset \Delta$ as $I_P := I_{P'}$, where P' is the unique conjugate of P containing B.

- For a parabolic $P \subset G_k$, write $\Delta^P := \Delta \setminus I_P$.
- For all $\alpha \in \Phi$, choose an isomorphism $u_\alpha \colon \mathbb{G}_{\mathrm{a}} \to U_\alpha$ so that $(u_\alpha)_{\alpha \in \Phi}$ is a realization in the sense of [Spr98, Section 8.1.4]. In particular, we have
$$tu_\alpha(x)t^{-1} = u_\alpha(\alpha(t)x), \quad \forall x \in \mathbb{G}_{\mathrm{a}},\ \forall t \in T.$$
- Let $\phi_\alpha \colon \mathrm{SL}_2 \to G$ denote the map attached to α as in [Spr98, Proof of Lemma 9.2.2]. It satisfies
$$\phi_\alpha\left(\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}\right) = u_\alpha(x), \quad \phi_\alpha\left(\begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix}\right) = u_{-\alpha}(x).$$
- Fix a B-representation (V, ρ). For $j \in \mathbb{Z}$ and $\alpha \in \Phi$, we define a map $E_\alpha^{(j)} \colon V \to V$ as follows. Let $V = \bigoplus_{\nu \in X^*(T)} V_\nu$ be the weight decomposition of V. For $v \in V_\nu$, we can write uniquely
$$u_\alpha(x)v = \sum_{j \geq 0} x^j E_\alpha^{(j)}(v), \quad \forall x \in \mathbb{G}_{\mathrm{a}},$$
for elements $E_\alpha^{(j)}(v) \in V_{\nu + j\alpha}$ (see [IK21, Lemma 3.3.1]). Extend $E_\alpha^{(j)}$ by additivity to a map $V \to V$. For $j < 0$, put $E_\alpha^{(j)} = 0$.

2.2 The stack of G-zips

We recall some facts about the stack of G-zips of Pink–Wedhorn–Ziegler in [PWZ11].

2.2.1 Definitions

Let G be a connected reductive group over $\mathbb{F}_q$. In this paper, a zip datum is a tuple $\mathcal{Z} := (G, P, L, Q, M)$ consisting of the following objects:

(i) $P \subset G_k$ and $Q \subset G_k$ are parabolic subgroups of G_k.

(ii) $L \subset P$ and $M \subset Q$ are Levi subgroups such that $L^{(q)} = M$.

For an algebraic group H, denote by $R_{\mathrm{u}}(H)$ the unipotent radical of H. If $P' \subset G_k$ is a parabolic subgroup with Levi subgroup $L' \subset P'$, any $x \in P'$ can be written uniquely as $x = \overline{x}u$ with $\overline{x} \in L'$ and $u \in R_{\mathrm{u}}(P')$. We denote by $\theta_{L'}^{P'} \colon P' \to L'$ the map $x \mapsto \overline{x}$. Since $M = L^{(q)}$, we have a Frobenius isogeny $\varphi \colon L \to M$. Put
$$E := \left\{ (x, y) \in P \times Q \;\middle|\; \varphi\left(\theta_L^P(x)\right) = \theta_M^Q(y) \right\}.$$
Equivalently, E is the subgroup of $P \times Q$ generated by $R_{\mathrm{u}}(P) \times R_{\mathrm{u}}(Q)$ and elements of the form $(a, \varphi(a))$ with $a \in L$. Let $G \times G$ act on G by $(a, b) \cdot g := agb^{-1}$, and let E act on G by restricting this action to E. The stack of G-zips of type $\mathcal{Z}$ (see [PWZ11, PWZ15]) can be defined as the quotient stack
$$G\text{-}\mathtt{Zip}^{\mathcal{Z}} = [E \backslash G_k].$$

2.2.2 Cocharacter datum

A *cocharacter datum* is a pair (G, μ), where G is a reductive connected group over $\mathbb{F}_q$ and $\mu \colon \mathbb{G}_{\mathrm{m},k} \to G_k$ is a cocharacter. One can attach to (G, μ) a zip datum $\mathcal{Z}_\mu$, defined as follows. First, denote by $P_+(\mu)$ (resp. $P_-(\mu)$) the unique parabolic subgroup of G_k such that $P_+(\mu)(k)$ (resp. $P_-(\mu)(k)$) consists of the elements $g \in G(k)$ satisfying that the map
$$\mathbb{G}_{\mathrm{m},k} \longrightarrow G_k, \quad t \longmapsto \mu(t)g\mu(t)^{-1} \quad \left(\text{resp. } t \longmapsto \mu(t)^{-1}g\mu(t)\right)$$
extends to a morphism of varieties $\mathbb{A}^1_k \to G_k$. We obtain a pair of parabolics $(P_+(\mu), P_-(\mu))$ in G_k whose intersection $P_+(\mu) \cap P_-(\mu) = L(\mu)$ is the centralizer of μ (it is a common Levi subgroup of $P_+(\mu)$ and $P_-(\mu)$). Set $P := P_-(\mu)$, $Q := (P_+(\mu))^{(q)}$, $L := L(\mu)$ and $M := (L(\mu))^{(q)}$. The tuple $\mathcal{Z}_\mu := (G, P, L, Q, M)$ is a zip datum, which we call the zip datum attached to the cocharacter datum (G, μ). We write simply $G\text{-}\mathtt{Zip}^{\mu}$ for $G\text{-}\mathtt{Zip}^{\mathcal{Z}_\mu}$. We always consider zip data of this form.

Remark 2.1. A general zip datum (G, P, L, Q, M) is of the form $\mathcal{Z}_\mu$ for a cocharacter $\mu \colon \mathbb{G}_{\mathrm{m},k} \to G_k$ if and only if $\sigma(P)$ and Q are opposite parabolic subgroups with common Levi $M = \sigma(L)$.

Remark 2.2. If μ is defined over $\mathbb{F}_q$, then so are P and Q. In this case, we have $L = M$, and P and Q are opposite parabolic subgroups with common Levi subgroup L.

2.2.3 Frames

Let $\mathcal{Z} = (G, P, Q, L, M)$ be a zip datum. In this paper, a frame for $\mathcal{Z}$ is a triple (B, T, z), where (B, T) is a Borel pair of G_k defined over $\mathbb{F}_q$ such that

(i) one has the inclusion $B \subset P$;

(ii) $z \in W$ is an element satisfying the conditions

$$^zB \subset Q \quad \text{and} \quad B \cap M = {}^zB \cap M.$$

We put $B_M := B \cap M$. Other papers (see [PWZ11, PWZ15, KW18]) use the convention $B \subset Q$ instead of $B \subset P$. A frame (as defined here) may not always exist. However, if (G, μ) is a cocharacter datum and $\mathcal{Z}_\mu$ is the associated zip datum by Section 2.2.2, then there exists a $G(k)$-conjugate $\mu' = \mathrm{ad}(g) \circ \mu$ (with $g \in G(k)$) such that $\mathcal{Z}_{\mu'}$ admits a frame, by Lemma 2.2.3 below. Hence, it is harmless to assume that a frame exists, and we only consider zip data that admit frames. With respect to the Borel pair (B, T), we define subsets I, J, Δ^P of Δ as follows:

$$I := I_P, \quad J := I_Q, \quad \Delta^P = \Delta \setminus I.$$

Lemma 2.3 (*cf.* [GK19b, Lemma 2.3.4])**.** *Let $\mu\colon \mathbb{G}_{\mathrm{m},k} \to G_k$ be a cocharacter, and let $\mathcal{Z}_\mu$ be the attached zip datum. Assume that (B, T) is a Borel pair defined over $\mathbb{F}_q$ such that $B \subset P$. Define the element*

$$z := w_0 w_{0,J} = \sigma(w_{0,I}) w_0.$$

Then (B, T, z) is a frame for $\mathcal{Z}_\mu$.

2.2.4 Parametrization of the E-orbits in G

By [PWZ11, Proposition 7.1], there are finitely many E-orbits in G. The E-orbits are smooth and locally closed in G, and the Zariski closure of an E-orbit is a union of E-orbits. We review the parametrization of E-orbits following [PWZ11]. For $w \in W$, fix a representative $\dot{w} \in N_G(T)$ such that $(w_1 w_2)^{\cdot} = \dot{w}_1 \dot{w}_2$ whenever $\ell(w_1 w_2) = \ell(w_1) + \ell(w_2)$ (this is possible by choosing a Chevalley system; see [ABD+11, Section XXIII.6]). For $w \in W$, define G_w as the E-orbit of $\dot{w}\dot{z}^{-1}$. If no confusion occurs, we write w instead of $\dot{w}$. For $w, w' \in {}^IW$, write $w' \preccurlyeq w$ if there exists a $w_1 \in W_I$ such that $w' \leq w_1 w \sigma(w_1)^{-1}$. This defines a partial order on ${}^IW$ (see [PWZ11, Corollary 6.3]).

Theorem 2.4 (*cf.* [PWZ11, Theorems 7.5, 11.2, 11.3 and 11.5])**.** *We have two bijections:*

$$^IW \longrightarrow \{E\text{-orbits in } G_k\}, \quad w \longmapsto G_w,$$
$$W^J \longrightarrow \{E\text{-orbits in } G_k\}, \quad w \longmapsto G_w.$$

For $w \in {}^IW \cup W^J$, one has $\dim(G_w) = \ell(w) + \dim(P)$, and the Zariski closure of G_w is

$$\overline{G}_w = \bigsqcup_{w' \in {}^IW,\ w' \preccurlyeq w} G_{w'}$$

for $w \in {}^IW$ and

$$\overline{G}_w = \bigsqcup_{w' \in W^J,\ w' \preccurlyeq w} G_{w'}$$

for $w \in W^J$.

In particular, there is a unique open E-orbit $U_{\mathcal{Z}} \subset G$ corresponding to the longest elements $w_{0,I} w_0 \in {}^IW$ via (2.4) and to $w_0 w_{0,J} \in W^J$ via (2.4). The E-orbit $U_{\mathcal{Z}}$ is dense in G. If $\mathcal{Z} = \mathcal{Z}_\mu$ (see Section 2.2.2), write $U_\mu = U_{\mathcal{Z}_\mu}$. In this case, we can choose $z = w_0 w_{0,J} = \sigma(w_{0,I}) w_0$ (see Lemma 2.3); hence (2.4) shows that $1 \in U_\mu$. We put $\mathcal{U}_\mu := [E \backslash U_\mu]$, which we call the μ-ordinary locus.

2.3 Vector bundles on the stack of G

2.3.1 Representation theory

For an algebraic group G over a field K, denote by $\mathrm{Rep}(G)$ the category of algebraic representations of G on finite-dimensional K-vector spaces. We denote a representation $\rho\colon G \to \mathrm{GL}_K(V)$ by (V,ρ), or sometimes simply ρ or V. For an algebraic group G over $\mathbb{F}_q$, a G_k-representation (V,ρ) and an integer m, we denote by $(V^{[m]},\rho^{[m]})$ the representation such that $V^{[m]} = V$ and

$$\rho^{[m]}\colon G_k \xrightarrow{\varphi^m} G_k \xrightarrow{\rho} \mathrm{GL}(V).$$

Let H be a split connected reductive K-group, and choose a Borel pair (B_H,T) defined over K. If K has characteristic zero, $\mathrm{Rep}(H)$ is semisimple. In characteristic p, however, this is no longer true in general. For $\lambda \in X_+^*(T)$, let $\mathcal{L}_\lambda$ be the line bundle attached to λ on the flag variety H/B_H by the usual associated sheaf construction (see [Jan03, Section 5.8]). Define an H-representation $V_H(\lambda)$ by

$$V_H(\lambda) := H^0\left(H/B_H, \mathcal{L}_\lambda\right).$$

In other words, one has $V_H(\lambda) = \mathrm{Ind}_{B_H}^H \lambda$. The representation $V_H(\lambda)$ is of highest weight λ. If $\mathrm{char}(K) = 0$, the representation $V_H(\lambda)$ is irreducible. We view elements of $V_H(\lambda)$ as regular maps $f\colon H \to \mathbb{A}^1$ satisfying

$$f(hb) = \lambda\left(b^{-1}\right) f(h), \quad \forall h \in H,\ \forall b \in B_H.$$

For dominant characters λ, λ', there is a natural surjective map

$$V_H(\lambda) \otimes V_H(\lambda') \longrightarrow V_H(\lambda + \lambda').$$

In the description given by (2.3.1), this map is $f \otimes f' \mapsto ff'$ (for $f \in V_H(\lambda)$, $f' \in V_H(\lambda')$). Denote by $W_H := W(H,T)$ the Weyl group and by $w_{0,H} \in W_H$ the longest element. Then $V_H(\lambda)$ has a unique B_H-stable line, which is a weight space for the weight $w_{0,H}\lambda$.

2.3.2 Vector bundles on quotient stacks

For an algebraic stack $\mathcal{X}$, write $\mathfrak{VB}(\mathcal{X})$ for the category of vector bundles on $\mathcal{X}$. Let X be a k-scheme and H an affine k-group scheme acting on X. If $\rho\colon H \to \mathrm{GL}(V)$ is an algebraic representation of H, it gives rise to a vector bundle $\mathcal{V}_{H,X}(\rho)$ on the stack $[H\backslash X]$. This vector bundle can be defined geometrically as $[H\backslash(X \times_k V)]$, where H acts diagonally on $X \times_k V$. We obtain a functor

$$\mathcal{V}_{H,X}\colon \mathrm{Rep}(H) \longrightarrow \mathfrak{VB}([H\backslash X]).$$

Similarly to the usual associated sheaf construction, see [Jan03, Section 5.8, Equation (1)], the global sections of $\mathcal{V}_{H,X}(\rho))$ are given by

$$H^0([H\backslash X], \mathcal{V}_{H,X}(\rho)) = \{f\colon X \to V \mid f(h\cdot x) = \rho(h)f(x),\ \forall h \in H,\ \forall x \in X\},$$

where $f\colon X \to V$ is a morphism of k-schemes, and V is viewed as an affine space over k.

2.3.3 Vector bundles on G-$\mathtt{Zip}^{\mu}$

Fix a cocharacter datum (G,μ). Let $\mathcal{Z} = (G,P,L,Q,M)$ be the attached zip datum. Fix a frame (B,T) as in Section 2.2.3. By (2.3.2), we have a functor $\mathcal{V}_{E,G}\colon \mathrm{Rep}(E) \to \mathfrak{VB}(G\text{-}\mathtt{Zip}^{\mu})$, which we simply denote by $\mathcal{V}$. For $(V,\rho) \in \mathrm{Rep}(E)$, the global sections of $\mathcal{V}(\rho)$ are

$$H^0(G\text{-}\mathtt{Zip}^{\mu}, \mathcal{V}(\rho)) = \{f\colon G_k \to V \mid f(\epsilon\cdot g) = \rho(\epsilon)f(g),\ \forall \epsilon \in E,\ \forall g \in G_k\}.$$

Since G admits an open dense E-orbit (see the discussion below Theorem 2.4), the space $H^0(G\text{-}\mathtt{Zip}^{\mu}, \mathcal{V}(\rho))$ is finite-dimensional (see [Kos19, Lemma 1.2.1]). The first projection $p_1\colon E \to P$ induces a functor $p_1^*\colon \mathrm{Rep}(P) \to \mathrm{Rep}(E)$. If $(V,\rho) \in \mathrm{Rep}(P)$, we write again $\mathcal{V}(\rho)$ for $\mathcal{V}(p_1^*(\rho))$. In this paper, we only consider E-representations coming from P in this way. Let $\theta_L^P\colon P \to L$ be the natural projection modulo $R_{\mathrm{u}}(P)$, as in Section 2.2.1. It induces a fully faithful functor

$$(\theta_L^P)^*\colon \mathrm{Rep}(L) \longrightarrow \mathrm{Rep}(P)$$

whose image is the full subcategory of $\mathrm{Rep}(P)$ of P-representations trivial on $R_{\mathrm{u}}(P)$. Hence, we view $\mathrm{Rep}(L)$ as a full subcategory of $\mathrm{Rep}(P)$. If $(V,\rho) \in \mathrm{Rep}(L)$, write again $\mathcal{V}(\rho) := \mathcal{V}((\theta_L^P)^*\rho)$. For $\lambda \in X^*(T)$, write $B_L := B \cap L$, and define an L-representation $(V_I(\lambda), \rho_{I,\lambda})$ as follows:

$$V_I(\lambda) = \mathrm{Ind}_{B_L}^{L} \lambda, \quad \rho_{I,\lambda} \colon L \longrightarrow \mathrm{GL}(V_I(\lambda)).$$

This is the representation defined in (2.3.1) for $H = L$ and $B_H = B_L$. Let $\mathcal{V}_I(\lambda)$ be the vector bundle on G-$\mathtt{Zip}^{\mu}$ attached to $V_I(\lambda)$, and call it an *automorphic vector bundle* on G-$\mathtt{Zip}^{\mu}$ associated to λ. This terminology stems from Shimura varieties (see Section 2.6 below for further details). For $\lambda \in X^*(L)$, viewing λ as an element of $X^*(T)$ by restriction, the vector bundle $\mathcal{V}_I(\lambda)$ is a line bundle. Note that if $\lambda \in X^*(T)$ is not I-dominant, then $V_I(\lambda) = 0$ and thus $\mathcal{V}_I(\lambda) = 0$.

2.4 Global sections over G-$\mathtt{Zip}^{\mu}$

We review some results of [IK21] regarding the global sections of $\mathcal{V}(\rho)$ for a P-representation ρ. We start with sections over the open substack $\mathcal{U}_\mu \subset G$-$\mathtt{Zip}^{\mu}$. Recall that $\mathcal{U}_\mu = [E\backslash U_\mu]$ and $1 \in U_\mu$ (see Section 2.2.4). By (2.3.2), an element of $H^0(\mathcal{U}_\mu, \mathcal{V}(\rho))$ can be viewed as a map $h \colon G \to V$ satisfying $h(agb^{-1}) = \rho(a)h(g)$ for all $(a,b) \in E$ and all $g \in G$. Since the E-orbit of 1 is open dense in G, the map $h \mapsto h(1)$ is an injection

$$\mathrm{ev}_1 \colon H^0\left(\mathcal{U}_\mu, \mathcal{V}(\rho)\right) \longrightarrow V.$$

We give the image of this map. Let L_φ be the scheme-theoretic stabilizer subgroup of 1 in E. By definition, one has

$$L_\varphi = E \cap \{(x,x) \mid x \in G_k\},$$

which is a 0-dimensional algebraic group (in general nonsmooth). The first projection $E \to P$ induces a closed immersion $L_\varphi \to P$. Identify L_φ with its image, and view it as a subgroup of P. Denote by $L_0 \subset L$ the largest algebraic subgroup defined over $\mathbb{F}_q$. In other words,

$$L_0 = \bigcap_{n \geq 0} L^{(q^n)}.$$

Lemma 2.5 (*cf.* [KW18, Lemma 3.2.1])**.**

(1) *One has $L_\varphi \subset L$.*

(2) *The group L_φ can be written as a semidirect product $L_\varphi = L_\varphi^\circ \rtimes L_0(\mathbb{F}_q)$, where L_φ° is the identity component of L_φ. Furthermore, L_φ° is a finite unipotent algebraic group.*

(3) *Assume that P is defined over $\mathbb{F}_q$. Then $L_0 = L$ and $L_\varphi = L(\mathbb{F}_q)$, viewed as a constant algebraic group.*

Lemma 2.6 (*cf.* [IK21, Corollary 3.2.3])**.** *The map* (2.4) *induces an identification*

$$H^0\left(\mathcal{U}_\mu, \mathcal{V}(\rho)\right) = V^{L_\varphi}.$$

Here, the notation V^{L_φ} denotes the space of scheme-theoretic invariants, *i.e.*, the set of $v \in V$ such that for any k-algebra R, one has $\rho(x)v = v$ in $V \otimes_k R$ for all $x \in L_\varphi(R)$. We now consider the space of global sections over G-$\mathtt{Zip}^{\mu}$. Restriction of sections to $\mathcal{U}_\mu \subset G$-$\mathtt{Zip}^{\mu}$ induces an injective map $H^0(G\text{-}\mathtt{Zip}^{\mu}, \mathcal{V}(\rho)) \to H^0(\mathcal{U}_\mu, \mathcal{V}(\rho)) = V^{L_\varphi}$. For simplicity, we assume here that P is defined over $\mathbb{F}_q$ (for the general result, see [IK21, Theorem 3.4.1]). We will need the general version in the proof of Proposition 3.8, but in the simple setting when ρ is a character $L \to \mathbb{G}_{\mathrm{m}}$. For $\alpha \in \Phi$, choose a realization $(u_\alpha)_{\alpha \in \Phi}$ (see Section 2.1). Fix a P-representation (V,ρ), and let $V = \bigoplus_{\nu \in X^*(T)} V_\nu$ be its T-weight decomposition. Define the Brylinski–Kostant filtration (*cf.* [XZ19, Equation (3.3.2)]) indexed by $c \in \mathbb{R}$ on V_ν by

$$\mathrm{Fil}_c^\alpha V_\nu = \bigcap_{j > c} \mathrm{Ker}\left(E_\alpha^{(j)} \colon V_\nu \to V_{\nu + j\alpha}\right),$$

where the map E_α was defined in Section 2.1. For $\chi \in X^*(T)_{\mathbb{R}}$ and $\nu \in X^*(T)$, also set

$$\mathrm{Fil}_\chi^P V_\nu = \bigcap_{\alpha \in \Delta^P} \mathrm{Fil}_{\langle \chi, \alpha^\vee \rangle}^{-\alpha} V_\nu.$$

The Lang torsor morphism $\wp\colon T \to T$, $g \mapsto g\varphi(g)^{-1}$ induces isomorphisms

$$\wp^*\colon X^*(T)_{\mathbb{R}} \xrightarrow{\sim} X^*(T)_{\mathbb{R}}, \quad \lambda \longmapsto \lambda \circ \wp = \lambda - q\sigma^{-1}(\lambda),$$
$$\wp_*\colon X_*(T)_{\mathbb{R}} \xrightarrow{\sim} X_*(T)_{\mathbb{R}}, \quad \delta \longmapsto \wp \circ \delta = \delta - q\sigma(\delta).$$

Theorem 2.7 (*cf.* [IK21, Corollary 3.4.2])**.** *Assume that P is defined over $\mathbb{F}_q$. For all $(V,\rho) \in \mathrm{Rep}(P)$, the map* ev_1 *induces an identification*

$$H^0(G\text{-}\mathtt{Zip}^\mu, \mathcal{V}(\rho)) = V^{L(\mathbb{F}_q)} \cap \bigoplus_{\nu \in X^*(T)} \mathrm{Fil}^P_{\wp^{*-1}(\nu)} V_\nu.$$

In the general case of an arbitrary parabolic P, $V^{L(\mathbb{F}_q)}$ is replaced by V^{L_φ} and $\mathrm{Fil}^\alpha_c V_\nu$ is replaced by a generalized Brylinski–Kostant filtration (see [IK21, Theorem 3.4.1]). In the special case when ρ is trivial on $R_{\mathrm{u}}(P)$, Theorem 2.7 simplifies greatly. Set $\delta_\alpha := \wp_*^{-1}(\alpha^\vee)$, and define a subspace $V^{\Delta^P}_{\geq 0} \subset V$ by

$$V^{\Delta^P}_{\geq 0} = \bigoplus_{\langle \nu, \delta_\alpha \rangle \geq 0,\ \forall \alpha \in \Delta^P} V_\nu.$$

If T is split over $\mathbb{F}_q$, then $\delta_\alpha = -\alpha^\vee/(q-1)$, and $V^{\Delta^P}_{\geq 0}$ is the direct sum of the weight spaces V_ν for those $\nu \in X^*(T)$ satisfying $\langle \nu, \alpha^\vee \rangle \leq 0$ for all $\alpha \in \Delta^P$.

Corollary 2.8. *Assume that P is defined over $\mathbb{F}_q$ and furthermore that $(V,\rho) \in \mathrm{Rep}(P)$ is trivial on $R_{\mathrm{u}}(P)$. Then one has*

$$H^0(G\text{-}\mathtt{Zip}^\mu, \mathcal{V}(\rho)) = V^{L(\mathbb{F}_q)} \cap V^{\Delta^P}_{\geq 0}.$$

2.5 The stack of G-zip flags

2.5.1 Definition

Let (G,μ) be a cocharacter datum with attached zip datum $\mathcal{Z}_\mu = (G,P,L,Q,M)$ (see Section 2.2.2). Fix a frame (B,T,z) with $z = \sigma(w_{0,I})w_0 = w_0 w_{0,J}$ (see Lemma 2.3). The stack of zip flags (see [GK19a, Definition 2.1.1]) is defined as

$$G\text{-}\mathtt{ZipFlag}^\mu = [E\backslash(G_k \times P/B)],$$

where the group E acts on the variety $G_k \times (P/B)$ by the rule $(a,b)\cdot(g,hB) := (agb^{-1}, ahB)$ for all $(a,b) \in E$ and all $(g,hB) \in G_k \times P/B$. The first projection $G_k \times P/B \to G_k$ is E-equivariant and yields a natural morphism of stacks

$$\pi\colon\ G\text{-}\mathtt{ZipFlag}^\mu \longrightarrow G\text{-}\mathtt{Zip}^\mu\,.$$

Set $E' := E \cap (B \times G_k)$. Then the injective map $G_k \to G_k \times P/B$, $g \mapsto (g,B)$ yields an isomorphism of stacks $[E'\backslash G_k] \simeq G\text{-}\mathtt{ZipFlag}^\mu$ (see [GK19a, Equation (2.1.5)]). We recall the stratification of $G\text{-}\mathtt{ZipFlag}^\mu$. First, define the Schubert stack as the quotient stack

$$\mathrm{Sbt} := [B\backslash G_k/B].$$

This stack is finite and smooth. Its topological space is isomorphic to W, endowed with the topology induced by the Bruhat order on W. This follows easily from the Bruhat decomposition of G. One can show that $E' \subset B \times {}^zB$. In particular, there is a natural projection map $[E'\backslash G_k] \to [B\backslash G_k/{}^zB]$. Composing with the isomorphism $[B\backslash G_k/{}^zB] \to [B\backslash G_k/B]$ induced by $G_k \to G_k$, $g \mapsto gz$, we obtain a smooth, surjective map

$$\psi\colon\ G\text{-}\mathtt{ZipFlag}^\mu \longrightarrow \mathrm{Sbt}\,.$$

For $w \in W$, put $\mathrm{Sbt}_w := [B\backslash BwB/B]$; it is a locally closed substack of Sbt. The *flag strata* of $G\text{-}\mathtt{ZipFlag}^\mu$ are defined as fibers of the map ψ. They are locally closed substacks (endowed with the reduced structure). Concretely, let $w \in W$, and put

$$F_w := B\left(wz^{-1}\right){}^zB = BwBz^{-1},$$

which is the $B \times {}^zB$-orbit of wz^{-1}. The set F_w is locally closed in G_k, and one has $\dim(F_w) = \ell(w) + \dim(B)$. Then, via the isomorphism $G\text{-}\mathtt{ZipFlag}^\mu \simeq [E'\backslash G_k]$, the flag strata of $G\text{-}\mathtt{ZipFlag}^\mu$ are the locally closed substacks

$$\mathcal{F}_w := [E'\backslash F_w], \quad w \in W.$$

The set $F_{w_0} \subset G_k$ is open in G_k, and similarly the stratum $\mathcal{F}_{w_0}$ is open in $G\text{-}\mathtt{ZipFlag}^\mu$. The $B \times {}^zB$-orbits of codimension 1 are $F_{s_\alpha w_0}$ for $\alpha \in \Delta$. The Zariski closure $\overline{F}_w$ is normal (see [RR85, Theorem 3]) and coincides with $\bigcup_{w' \leq w} F_{w'}$, where $\leq$ is the Bruhat order of W.

2.5.2 Vector bundles on G-$\mathtt{ZipFlag}^{\mu}$

Let $\rho\colon B \to \mathrm{GL}(V)$ be an algebraic representation, and view ρ as a representation of E' via the first projection $E' \to B$. Via the isomorphism $G\text{-}\mathtt{ZipFlag}^{\mu} \simeq [E'\backslash G_k]$, we obtain a vector bundle $\mathcal{V}_{\mathsf{flag}}$ on $G\text{-}\mathtt{ZipFlag}^{\mu}$. Let $(V,\rho) \in \mathrm{Rep}(P)$, and let $\mathcal{V}(\rho)$ be the attached vector bundle on $G\text{-}\mathtt{Zip}^{\mu}$. Then one has

$$\pi^*(\mathcal{V}(\rho)) = \mathcal{V}_{\mathsf{flag}}(\rho|_B).$$

Note that the rank of $\mathcal{V}_{\mathsf{flag}}(\rho)$ is the dimension of ρ. In particular, if $\lambda \in X^*(B)$, then $\mathcal{V}_{\mathsf{flag}}(\lambda)$ is a line bundle. For $(V,\rho) \in \mathrm{Rep}(B)$, consider the P-representation $\mathrm{Ind}_B^P(\rho)$ defined by

$$\mathrm{Ind}_B^P(\rho) = \left\{ f\colon P \to V \;\middle|\; f(xb) = \rho\left(b^{-1}\right) f(x),\ \forall b \in B,\ \forall x \in P \right\}.$$

For $y \in P$ and $f \in \mathrm{Ind}_B^P(\rho)$, the element $y \cdot f$ is the function $x \mapsto f(y^{-1}x)$.

Proposition 2.9 (*cf.* [IK24, Proposition 3.2.1])**.** *For $(V,\rho) \in \mathrm{Rep}(B)$, we have the identification $\pi_*(\mathcal{V}_{\mathsf{flag}}(\rho)) = \mathcal{V}(\mathrm{Ind}_B^P(\rho))$. In particular $\pi_*(\mathcal{V}_{\mathsf{flag}}(\rho))$ is a vector bundle on $G\text{-}\mathtt{Zip}^{\mu}$.*

In particular, if ρ is a character $\lambda \in X^*(T)$, then $\mathcal{V}_{\mathsf{flag}}(\lambda)$ is a line bundle and one has

$$\pi_*\left(\mathcal{V}_{\mathsf{flag}}(\lambda)\right) = \mathcal{V}_I(\lambda),$$

where the vector bundle $\mathcal{V}_I(\lambda)$ was defined in Section 2.3.3. Hence, we have

$$H^0(G\text{-}\mathtt{Zip}^{\mu}, \mathcal{V}_I(\lambda)) = H^0\left(G\text{-}\mathtt{ZipFlag}^{\mu}, \mathcal{V}_{\mathsf{flag}}(\lambda)\right).$$

If $f\colon G_k \to k$ is a section of the right-hand side of (2.5.2), then the corresponding function $f_I\colon G_k \to V_I(\lambda)$ on the left-hand side of (2.5.2) is given by

$$(f_I(g))(x) = f\left(\left(x^{-1}, \varphi(x)^{-1}\right) \cdot g\right) = f\left(x^{-1} g \varphi(x)\right)$$

for all $g \in G_k$ and $x \in L$, by the construction of the identification. Also note that the line bundles $\mathcal{V}_{\mathsf{flag}}(\lambda)$ satisfy the following identity:

$$\mathcal{V}_{\mathsf{flag}}(\lambda + \lambda') = \mathcal{V}_{\mathsf{flag}}(\lambda) \otimes \mathcal{V}_{\mathsf{flag}}(\lambda'), \quad \forall \lambda, \lambda' \in X^*(T).$$

We can also define vector bundles on the stack Sbt as in [IK24, Section 4]. For our purpose, it is enough to define line bundles on Sbt. Using (2.3.2), we can attach to each $(\chi_1, \chi_2) \in X^*(T) \times X^*(T)$ a line bundle $\mathcal{V}_{\mathrm{Sbt}}(\chi_1, \chi_2)$ on Sbt. One has

$$\psi^* \mathcal{V}_{\mathrm{Sbt}}(\chi_1, \chi_2) = \mathcal{V}_{\mathsf{flag}}(\chi_1 + (z\chi_2) \circ \varphi) = \mathcal{V}_{\mathsf{flag}}\left(\chi_1 + q\sigma^{-1}(z\chi_2)\right).$$

2.5.3 Partial Hasse invariants

We recall some results of [IK24]. By [GK19a, Theorem 2.2.1(a)], the line bundle $\mathcal{V}_{\mathrm{Sbt}}(\chi_1, \chi_2)$ admits a nonzero section over Sbt_{w_0} if and only if $\chi_1 = -w_0\chi_2$. If this condition is satisfied, $H^0(\mathrm{Sbt}_{w_0}, \mathcal{V}_{\mathrm{Sbt}}(\chi_1, \chi_2))$ is 1-dimensional. For $\chi \in X^*(T)$, let h_χ be any nonzero element

$$h_\chi \in H^0\left(\mathrm{Sbt}_{w_0}, \mathcal{V}_{\mathrm{Sbt}}(-w_0\chi, \chi)\right).$$

By [GK19a, Theorem 2.2.1(c)], h_χ extends to Sbt if and only if χ is a dominant character. Using (2.5.2) and $z = \sigma(w_{0,I})w_0$, we obtain a section

$$\mathrm{Ha}_\chi := \psi^*(h_\chi) \in H^0\left(\mathcal{F}_{w_0}, \mathcal{V}_{\mathsf{flag}}\left(-w_0\chi + q w_{0,I} w_0\left(\sigma^{-1}\chi\right)\right)\right),$$

and for $\chi \in X^*_+(T)$ the section Ha_χ extends to $G\text{-}\mathtt{ZipFlag}^{\mu}$. In particular, let $\alpha \in \Delta$, and suppose χ_α is a character satisfying

$$\begin{cases} \langle \chi_\alpha, \alpha^\vee \rangle > 0, \\ \langle \chi_\alpha, \beta^\vee \rangle = 0 \qquad \text{for all } \beta \in \Delta \setminus \{\alpha\}. \end{cases}$$

In this case, the section h_{χ_α} vanishes exactly on the codimension 1 stratum $\overline{\mathrm{Sbt}}_{w_0 s_\alpha}$. Similarly, the section $\mathrm{Ha}_{\chi_\alpha}$ cuts out the Zariski closure of the codimension 1 stratum $\mathcal{F}_{w_0 s_\alpha}$.

Definition 2.10. For $\alpha \in \Delta$ and χ_α satisfying (2.5.3), we call the section $\mathrm{Ha}_{\chi_\alpha}$ a partial Hasse invariant for the stratum $\mathcal{F}_{w_0 s_\alpha}$.

2.6 Shimura varieties and G-zips

We explain the connection between the stack of G-zips and Shimura varieties. Let $(\mathbf{G}, \mathbf{X})$ be a Shimura datum; see [Del79, Section 2.1.1]. Write $\mathbf{E} = E(\mathbf{G}, \mathbf{X})$ for the reflex field of $(\mathbf{G}, \mathbf{X})$ and $\mathcal{O}_{\mathbf{E}}$ for its ring of integers. Given an open compact subgroup $K \subset \mathbf{G}(\mathbf{A}_f)$, write $\mathrm{Sh}(\mathbf{G}, \mathbf{X})_K$ for Deligne's canonical model at level K over $\mathbf{E}$ (see [Del79]). For $K \subset \mathbf{G}(\mathbb{A}_f)$ small enough, $\mathrm{Sh}(\mathbf{G}, \mathbf{X})_K$ is a smooth, quasi-projective scheme over $\mathbf{E}$. Every inclusion $K' \subset K$ induces a finite étale projection $\pi_{K'/K} \colon \mathrm{Sh}(\mathbf{G}, \mathbf{X})_{K'} \to \mathrm{Sh}(\mathbf{G}, \mathbf{X})_K$.

Fix a prime number p, and assume that the level K is of the form $K = K_p K^p$, where $K_p \subset \mathbf{G}(\mathbb{Q}_p)$ is a hyperspecial subgroup and $K^p \subset \mathbf{G}(\mathbb{A}_f^p)$ is an open compact subgroup. Then one has $K_p = \mathscr{G}(\mathbb{Z}_p)$, where $\mathscr{G}$ is a reductive group over $\mathbb{Z}_p$ such that $\mathscr{G} \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \simeq \mathbf{G}_{\mathbb{Q}_p}$ and $\mathscr{G} \otimes_{\mathbb{Z}_p} \mathbb{F}_p$ is connected.

We assume that $(\mathbf{G}, \mathbf{X})$ is of abelian type. For any place v above p in $\mathbf{E}$, Kisin [Kis10] and Vasiu [Vas99] constructed a smooth, canonical model $\mathscr{S}_K$ of $\mathrm{Sh}(\mathbf{G}, \mathbf{X})_K$ over $\mathcal{O}_{\mathbf{E}_v}$. Let $\kappa(v)$ denote the residue field of $\mathcal{O}_{\mathbf{E}_v}$, and let $\overline{\mathbb{F}}_p$ be an algebraic closure of $\kappa(v)$. Set $S_K := \mathscr{S}_K \otimes_{\mathcal{O}_{\mathbf{E}_v}} \overline{\mathbb{F}}_p$. We can take a representative $\mu \in \{\mu\}$ defined over $\mathbf{E}_v$, by [Kot84, Lemma (1.1.3)(a)]. We can also assume that μ extends to $\mu \colon \mathbb{G}_{\mathrm{m},\mathcal{O}_{\mathbf{E}_v}} \to \mathscr{G}_{\mathcal{O}_{\mathbf{E}_v}}$ (see [Kim18, Corollary 3.3.11]). It gives rise to a parabolic subgroup $\mathscr{P} \subset \mathscr{G}_{\mathcal{O}_{\mathbf{E}_v}}$, with Levi subgroup $\mathscr{L}$ equal to the centralizer of μ. As explained in [IK21, Section 2.5], we can assume (after possibly twisting μ) that there is a Borel pair $(\mathscr{B}, \mathscr{T})$ over $\mathbb{Z}_p$ in $\mathscr{G}$ such that $\mathscr{B}_{\mathcal{O}_{\mathbf{E}_v}} \subset \mathscr{P}$. Let G, P, B, T denote the special fibers of $\mathscr{G}, \mathscr{P}, \mathscr{B}, \mathscr{T}$, respectively. By slight abuse of notation, we denote again by μ its mod p reduction $\mu \colon \mathbb{G}_{\mathrm{m},k} \to G_k$.

We define a quotient $\mathbf{G}^c$ of $\mathbf{G}$ by a subtorus of the center of $\mathbf{G}$ as in [IKY23, Section 2.3]. We note that $\mathbf{G}^c = \mathbf{G}$ if $(\mathbf{G}, \mathbf{X})$ is of Hodge type by [IKY23, Remark 2.6]. Let G^c be the quotients of G determined by $\mathbf{G}^c$. Let $\mu^c \colon \mathbb{G}_{\mathrm{m},k} \to G^c_k$ be the cocharacter induced by μ. Then (G^c, μ^c) is a cocharacter datum, and it yields a zip datum as in Section 2.2.2, where $q = p$. By [Zha18, Section 4.1] and [IKY23, Section 3.5], there exists a natural smooth morphism

$$\zeta \colon S_K \longrightarrow G^c\text{-}\mathtt{Zip}^{\mu^c}.$$

This map is also surjective by [SYZ21, Corollary 3.5.3(1)].

Let T^c be the maximal torus of G^c determined by T. For $\lambda \in X^*_{+,I}(T^c)$, we have a vector bundle $\mathcal{V}_I(\lambda)$ on $G^c\text{-}\mathtt{Zip}^{\mu^c}$ as in Section 2.3.3. We denote the pullback of $\mathcal{V}_I(\lambda)$ under ζ by the same symbol. The vector bundle $\mathcal{V}_I(\lambda)$ on S_K is called *the automorphic vector bundle of weight* λ.

The flag space of the Siegel modular variety $\mathcal{A}_n$ was first introduced by Ekedahl–van der Geer in [EvdG09]. It parametrizes pairs $(\underline{A}, \mathcal{F}_\bullet)$, where $\underline{A} = (A, \lambda) \in \mathcal{A}_n$ is a principally polarized abelian variety of relative dimension n and $\mathcal{F}_\bullet \subset H^1_{\mathrm{dR}}(A)$ is a full symplectic flag refining the Hodge filtration of $H^1_{\mathrm{dR}}(A)$. In general, the flag space $\mathrm{Flag}(S_K)$ of S_K was defined in [GK19a, Section 9.1] as the fiber product

$$\begin{array}{ccc} \mathrm{Flag}(S_K) & \xrightarrow{\zeta_{\mathsf{flag}}} & G\text{-}\mathtt{ZipFlag}^{\mu} \\ {\scriptstyle \pi_K}\downarrow & & \downarrow{\scriptstyle \pi} \\ S_K & \xrightarrow[\zeta]{} & G\text{-}\mathtt{Zip}^{\mu}. \end{array}$$

The stratification $(\mathcal{F}_w)_{w \in W}$ on $G\text{-}\mathtt{ZipFlag}^{\mu}$ induces by pullback via ζ_{flag} a stratification $(\mathrm{Flag}(S_K)_w)_{w \in W}$ of $\mathrm{Flag}(S_K)$ by locally closed, smooth subschemes. Moreover, we obtain a line bundle $\mathcal{V}_{\mathsf{flag}}(\lambda)$ on $\mathrm{Flag}(S_K)$. Since ζ is smooth, pullback and pushforward commute, so we have $\pi_{K,*}(\mathcal{V}_{\mathsf{flag}}(\lambda)) = \mathcal{V}_I(\lambda)$. In particular, the space of automorphic forms $H^0(S_K, \mathcal{V}_I(\lambda))$ can be identified with $H^0(\mathrm{Flag}(S_K), \mathcal{V}_{\mathsf{flag}}(\lambda))$.

3.5 Norm of the highest-weight vector

Recall that we always have $1 \in U_\mu$, where $U_\mu \subset G_k$ is the unique open E-orbit. Recall the definition of the finite subgroup $L_\varphi \subset L$ given in (2.4). Put $N_\varphi = |L_0(\mathbb{F}_q)|q^m$, where L_0 is the Levi subgroup defined in (2.4) and $m \geq 0$ is the smallest integer such that the finite unipotent group L°_φ is annihilated by φ^m. For $\lambda, \lambda' \in X^*(T)$ and $f \in V_I(\lambda)$, $f' \in V_I(\lambda')$, let $ff' \in V_I(\lambda + \lambda')$ be the image of $f \otimes f'$ by the map (2.3.1). For $\lambda \in X^*_{+,I}(T)$ and $f \in V_I(\lambda)$, define

$$\mathrm{Norm}_{L_\varphi}(f) := \left(\prod_{s \in L_0(\mathbb{F}_q)} s \cdot f \right)^{q^m} \in V_I(N_\varphi \lambda).$$

It is clear that $\mathrm{Norm}_{L_\varphi}(f)$ is L_φ-invariant. In particular, it gives rise to an element in $H^0(\mathcal{U}_\mu, \mathcal{V}(N_\varphi\lambda))$ by Lemma 2.6. In general, it is difficult to determine whether $\mathrm{Norm}_{L_\varphi}(f)$ extends to a global section. However, this is possible when f is a highest-weight vector, as we now explain.

Let $f_\lambda \in V_I(\lambda)$ be a nonzero element in the highest-weight line of $V_I(\lambda)$. The following result generalizes [Kos19, Theorem 2] (where P was assumed to be defined over $\mathbb{F}_p$; here we do not make this assumption). For $\alpha \in \Delta^P$, denote by r_α the smallest integer $r \geq 1$ such that $\sigma^r(\alpha) = \alpha$.

Proposition 3.9. *The section* $\mathrm{Norm}_{L_\varphi}(f_\lambda)$ *extends to a global section over* $G\text{-}\mathtt{Zip}^\mu$ *if and only if for all* $\alpha \in \Delta^P$*, the following holds:*

$$\sum_{w \in W_{L_0}(\mathbb{F}_q)} \sum_{i=0}^{r_\alpha - 1} q^{i+\ell(w)} \left\langle w\lambda, \sigma^i(\alpha^\vee) \right\rangle \leq 0.$$

Before giving the proof, we need to recall some facts from [IK21, Section 3.1]. First, we have

$$G_k \setminus U_\mu = \bigcup_{\alpha \in \Delta^P} Z_\alpha, \quad Z_\alpha = \overline{E \cdot s_\alpha},$$

where $E \cdot s_\alpha$ denotes the E-orbit of s_α and the bar denotes the Zariski closure. This follows easily from Theorem 2.4. For any $\alpha \in \Delta^P$, define an open subset

$$X_\alpha := G_k \setminus \bigcup_{\beta \in \Delta^P,\, \beta \neq \alpha} Z_\beta.$$

Then $U_\mu \subset X_\alpha$, and $X_\alpha \setminus U_\mu$ is irreducible. Choose a realization $(u_\alpha)_{\alpha \in \Phi}$, and let $\phi_\alpha \colon \mathrm{SL}_2 \to G$ be the map attached to α (see Section 2.1). Set $Y := E \times \mathbb{A}^1$ and $Y_0 := E \times \mathbb{G}_{\mathrm{m}}$. For $\alpha \in \Delta^P$, define $\psi_\alpha \colon Y \to G$ by

$$\psi_\alpha \colon ((x,y),t) \longmapsto x\phi_\alpha\left(A(t)\right) y^{-1}, \quad \text{where } A(t) = \begin{pmatrix} t & 1 \\ -1 & 0 \end{pmatrix} \in \mathrm{SL}_{2,k}\,.$$

It satisfies $\psi_\alpha((x,y),t) \in X_\alpha$ for all $((x,y),t) \in Y$ and $\psi_\alpha((x,y),t) \in U_\mu$ if and only if $t \neq 0$ (see [IK21, Proposition 3.1.4]).

We now prove Proposition 3.9. We use a similar argument to that in [Kos19, Theorem 3.5.3]. Set $\mathcal{U}'_\mu := \pi^{-1}(\mathcal{U}_\mu)$, where $\pi \colon G\text{-}\mathtt{ZipFlag}^\mu \to G\text{-}\mathtt{Zip}^\mu$ is the natural projection. One clearly has $\mathcal{U}'_\mu \simeq [E' \backslash U_\mu]$ via the isomorphism $G\text{-}\mathtt{ZipFlag}^\mu \simeq [E' \backslash G]$ explained in Section 2.5. We have an identification

$$H^0\left(\mathcal{U}_\mu, \mathcal{V}_I\left(N_\varphi\lambda\right)\right) = H^0\left(\mathcal{U}'_\mu, \mathcal{V}_{\mathsf{flag}}\left(N_\varphi\lambda\right)\right)$$

similarly to (2.5.2). In particular, we can view $\mathrm{Norm}_{L_\varphi}(f_\lambda)$ as a function $h \colon U_\mu \to \mathbb{A}^1$ satisfying the relation $h(axb^{-1}) = \lambda(a)h(x)$ for all $(a,b) \in E'$ and $x \in U_\mu$ (using (2.3.2)). Specifically, the function h is given by

$$h\left(x_1 x_2^{-1}\right) = \mathrm{Norm}_{L_\varphi}(f_\lambda)\left(\theta_L^P(x_1)^{-1}\right)$$

for all $(x_1, x_2) \in E$ using (2.5.2). The function h is well defined because $\mathrm{Norm}_{L_\varphi}(f_\lambda)$ is L_φ-invariant. Furthermore, $\mathrm{Norm}_{L_\varphi}(f_\lambda)$ extends to $G\text{-}\mathtt{Zip}^\mu$ if and only if h extends to a function $G \to \mathbb{A}^1$. By the strategy explained in [Kos19, Section 3.2] and in [IK21, Section 3.1], the function h extends to G if and only if for each $\alpha \in \Delta^P$, the function $h \circ \psi_\alpha \colon Y_0 \to \mathbb{A}^1$ extends to a function $Y \to \mathbb{A}^1$. It remains to compute the t-valuation of the function $h \circ \psi_\alpha$, viewed as an element of $R[t, \frac{1}{t}]$, where $R = k[E]$ is the ring of functions of E. Put

$$m_\alpha = \min\left\{m \geq 1 \mid \sigma^{-m}(\alpha) \notin I\right\}, \quad \alpha \in \Delta^P$$

and $t_\alpha = t^{-1}\alpha(\varphi(\delta_\alpha(t)))^{-1} = t\alpha(\delta_\alpha(t))^{-1} \in t^{\mathbb{Q}}$, where t is an indeterminate and $\delta_\alpha = \wp_*^{-1}(\alpha^\vee)$ as defined in Section 2.4. Set

$$u_{t,\alpha} = \prod_{i=1}^{m_\alpha - 1} \phi_{\sigma^{-i}(\alpha)}\left(\begin{pmatrix} 1 & -t_\alpha^{\frac{1}{q^i}} \\ 0 & 1 \end{pmatrix}\right),$$

where the product is taken in increasing order of indices. By the proof of [IK21, Proposition 3.1.4], for all $(x,y) \in E$ and $t \in \mathbb{G}_{\mathrm{m}}$, we can write $\psi_\alpha((x,y),t) = x_1 x_2^{-1}$ with $(x_1, x_2) \in E$ and

$$x_1 = x\phi_\alpha\left(\begin{pmatrix} 1 & 0 \\ -t^{-1} & 1 \end{pmatrix}\right)\delta_\alpha(t)u_{t,\alpha}.$$

By the definition of m_α, all the roots $\sigma^{-i}(\alpha)$ (for $1 \leq i \leq m_\alpha - 1$) appearing in the formula of $u_{t,\alpha}$ lie in I. Using (3.5), we deduce

$$\begin{aligned} h \circ \psi_\alpha((x,y),t) &= \mathrm{Norm}_{L_\varphi}(f_\lambda)\left(u_{t,\alpha}^{-1}\delta_\alpha(t)^{-1}\theta_L^P(x)^{-1}\right) \\ &= \left(\prod_{s \in L_0(\mathbb{F}_q)} f_\lambda\left(s u_{t,\alpha}^{-1}\delta_\alpha(t)^{-1}\theta_L^P(x)^{-1}\right)\right)^{q^m}. \end{aligned}$$

Consider the element $f_\lambda(su_{t,\alpha}^{-1}\delta_\alpha(t)^{-1}\theta_L^P(x)^{-1})$, which lies in $R[t^{\mathbb{Q}}]$. We can still speak of the t-valuation of this element, which is a rational number. Equivalently, to simplify notation, we change $\theta_L^P(x)^{-1}$ to a generic element $g \in L$, and we compute the t-valuation of $F_s(t,g) := f_\lambda(su_{t,\alpha}^{-1}\delta_\alpha(t)^{-1}g)$, viewed as an element of $k[L][t^{\mathbb{Q}}]$. Let $v_\alpha(s)$ be this valuation. We put $B_L^+ = B^+ \cap L$. Define a parabolic subgroup of L by $Q_0 := L_0 B_L^+$. It is clear that $u_{t,\alpha}$ lies in $R_{\mathrm{u}}(Q_0)$; thus for all $s \in L_0(\mathbb{F}_q)$, we have $su_{t,\alpha}^{-1}s^{-1} \in R_{\mathrm{u}}(Q_0)$. Since f_λ is invariant by $R_{\mathrm{u}}(B_L^+)$, we obtain $F_s(t,g) = f_\lambda(s\delta_\alpha(t)^{-1}g)$. Now, the rest of the proof is completely similar to that of [Kos19, Theorem 3.5.3]. We recall it briefly.

Let $B_{L_0} := B \cap L_0$ and $B_{L_0}^+ := B^+ \cap L_0$. If we change s to bs with $b \in B_{L_0}^+(\mathbb{F}_q)$, then $v_\alpha(bs) = v_\alpha(s)$. Indeed, this follows from $f_\lambda(bs\delta_\alpha(t)^{-1}g) = \lambda(b)^{-1}f_\lambda(s\delta_\alpha(t)^{-1}g)$ since f_λ is a B_L^+-eigenfunction. Similarly, we claim that $v_\alpha(sb) = v_\alpha(s)$ for all $b \in B_{L_0}^+(\mathbb{F}_q)$. By symmetry, it suffices to show $v_\alpha(sb) \geq v_\alpha(s)$. We can write $F_{sb}(t,g) = F_s(t,\Gamma(t)g)$, where

$$\Gamma(t) := \delta_\alpha(t) b \delta_\alpha(t)^{-1}.$$

We view Γ as a map $\Gamma\colon \mathrm{Spec}(k[t^{\mathbb{Q}}]) \to L$. Since $\alpha \in \Delta^P$, the cocharacter $\alpha^\vee$ is anti-L-dominant. It follows that for all $j \in \mathbb{Z}$, $\sigma^j(\alpha^\vee)$ is an anti-L_0-dominant quasi-cocharacter. It is easy to see that δ_α is explicitly given by the formula

$$\delta_\alpha = -\frac{1}{q^{r_\alpha}-1}\sum_{j=0}^{r_\alpha - 1} q^j \sigma^j\left(\alpha^\vee\right).$$

In particular, δ_α is L_0-dominant. We deduce that the function Γ extends to a map $\mathrm{Spec}(k[t^{\mathbb{Q}_{\geq 0}}]) \to L$. This follows simply from the fact that $\delta_\alpha(t)$ acts on the root space U_β (for $\beta \in \Phi$) by $t^{\langle \beta, \delta_\alpha\rangle}$, using (2.1). Write $F_s(t,g) = t^{v_\alpha(s)}F_{s,0}(t,g)$, where $F_{s,0}(t,g)$ is an element of $k[L][t^{\mathbb{Q}}]$ whose t-valuation is 0. Then $F_{sb}(t,g) = t^{v_\alpha(s)}F_{s,0}(t,\Gamma(t)g)$ and $F_{s,0}(t,\Gamma(t)g) \in k[L][t^{\mathbb{Q}_{\geq 0}}]$. Hence $v_\alpha(sb) \geq v_\alpha(s)$ as claimed.

Now, consider the Bruhat decomposition of $L_0(\mathbb{F}_q)$:

$$L_0(\mathbb{F}_q) = \bigsqcup_{w \in W_{L_0}(\mathbb{F}_q)} B_{L_0}^+(\mathbb{F}_q)\, w B_{L_0}^+(\mathbb{F}_q)$$

as in [Kos19, Lemma 3.4.4]. By [Kos19, Lemma 3.4.5], one has

$$\left|B_{L_0}^+(\mathbb{F}_q)\, w B_{L_0}^+(\mathbb{F}_q)\right| = |T(\mathbb{F}_q)|\, q^{\dim(R_{\mathrm{u}}(B_{L_0}))+\ell(w)}.$$

Thus, we can determine v_α completely from the values $v_\alpha(w)$ for $w \in W_{L_0}(\mathbb{F}_q)$. Similarly to [Kos19, Proposition 3.5.2], we have $v_\alpha(w) = \langle w\lambda, \delta_\alpha\rangle$. We deduce that the t-valuation of $h \circ \psi_\alpha((x,y),t)$ is

$$q^m \sum_{s \in L_0(\mathbb{F}_q)} v_\alpha(s) = q^m\,|T(\mathbb{F}_q)|\, q^{\dim(R_{\mathrm{u}}(B_{L_0}))} \sum_{w \in W_{L_0}(\mathbb{F}_q)} q^{\ell(w)}\langle w\lambda, \delta_\alpha\rangle.$$

The statement of Proposition 3.9 then follows by replacing δ_α by the expression in (3.5).

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